

One-Sample and Two-Sample Tests

A Rigorous Guide to Means, Proportions, Variances, Resampling, and High-Dimensional Outcomes

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Organization and clinical-trial setting

These two sessions move from familiar parametric tests to procedures whose calibration uses fewer distributional assumptions or more complex data structures. Every test is described through its null hypothesis, statistic, null distribution or approximation, procedure for calculating a p-value, and assumptions.

Session 1	One sample: a single treatment cohort versus a clinical benchmark. Tests for means, response proportions, and variances; asymptotic and exact procedures; bootstrap; nonparametric location tests; multivariate outcomes.
Session 2	Two samples: randomized treatment versus control. Independent and paired designs; tests for mean, proportion, and spread contrasts; asymptotic and exact procedures; randomization and permutation tests; bootstrap; rank tests; multivariate outcomes.

Let X_i be patient i 's change in systolic blood pressure from baseline to week 12; negative values indicate a reduction. In Session 1, a single-arm study may compare the mean change with a benchmark μ_0 . Let $R_i = 1$ if patient i meets a prespecified clinical response criterion, giving a binary endpoint. Let X_i also denote any continuous endpoint whose variability is of interest. In Session 2, let X_i and Y_j be outcomes from randomized treatment and control patients. If the same patients receive both conditions, analyze their within-patient differences.

1 General testing principles

1.1 Test, size, power, p-value

A test of $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ is a measurable function $\phi(X) \in [0, 1]$, interpreted as the conditional probability of rejection. Its size is $\sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\phi(X)]$; it has level at most α if this is no larger than α . Its power at $\theta \in \Theta_1$ is $\mathbb{E}_\theta[\phi(X)]$.

For a statistic T where large values favor an upper-tailed alternative, the p-value is $\mathbb{P}_{H_0}(T \geq T_{obs})$. For a lower-tailed alternative use $\mathbb{P}_{H_0}(T \leq T_{obs})$. For a two-sided alternative, use both tails. When the null distribution is symmetric and continuous, twice the smaller one-sided tail is a standard convention. For discrete or asymmetric statistics, two-sided p-values are not unique; name the convention. A p-value is not the probability that H_0 is true.

1.2 Exact, asymptotic, bootstrap, and randomization calibration

- **Exact model-based test:** the null distribution of the statistic is derived exactly under a sampling model, such as normal or binomial sampling.
- **Asymptotic test:** the null distribution is approximated as sample size grows, usually by a normal, t , chi-square, or F law. Its accuracy is a finite-sample question.
- **Bootstrap test:** resampling approximates a sampling distribution. To test a null, the bootstrap data-generating distribution must impose that null.
- **Permutation test:** labels are reassigned under an exchangeability null, usually equality of full distributions.
- **Randomization test:** treatment labels are reassigned according to the trial's actual randomization scheme. Under a sharp no-effect null, this can be finite-sample exact without a normal outcome model.

2 Session 1: one-sample tests

2.1 Targets and notation

Let $X_1, \dots, X_n$ be independent observations from a population with mean μ and variance σ^2 . Define

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

For a mean test, the null is typically $H_0 : \mu = \mu_0$. For variance, $H_0 : \sigma^2 = \sigma_0^2$. For a binary response $R_i \in \{0, 1\}$ with response probability p , the null is $H_0 : p = p_0$ and $K = \sum_i R_i$ is the number of responses.

2.2 One-sample mean

A. Large-sample (asymptotic) z test

Hypotheses. $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$, $\mu < \mu_0$, or $\mu \neq \mu_0$.

Statistic. If the population variance is known,

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

If it is unknown, use the studentized statistic

$$Z_S = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

Calibration and procedure. Under iid sampling with finite positive variance, $Z_S \Rightarrow N(0, 1)$ under H_0 . At level α , reject for $Z_S > z_{1-\alpha}$ (upper tail), $Z_S < z_\alpha$ (lower tail), or $|Z_S| > z_{1-\alpha/2}$ (two-sided). The corresponding p-values are $1 - \Phi(z_{obs})$, $\Phi(z_{obs})$, and $2\{1 - \Phi(|z_{obs}|)\}$. With known

σ , the CLT gives the same approximation; if the population is normal, the known-variance Z law is exact.

Assumptions and interpretation. Observations must represent independent sampling units (or dependence must be modeled). Finite variance is required for the usual CLT. Strong skewness, heavy tails, outliers, clustering, or small effective sample size can make the normal approximation poor. The test concerns the population mean, even if the sample median or a trimmed mean is also reported.

B. Exact one-sample Student t test

Model and hypotheses. Assume $X_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with unknown σ^2 . Test $H_0 : \mu = \mu_0$ against the same one- or two-sided alternatives.

Statistic and null law.

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0.$$

Indeed, $Z_0 = \sqrt{n}(\bar{X} - \mu_0)/\sigma \sim N(0, 1)$ and $V = (n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$ independently, so $T = Z_0/\sqrt{V/(n-1)}$.

Procedure. For an upper-tail alternative, reject if $T > t_{n-1, 1-\alpha}$ and report $1 - F_{t_{n-1}}(T_{\text{obs}})$. For a lower-tail alternative, use the lower tail. For a two-sided alternative, reject if $|T| > t_{n-1, 1-\alpha/2}$ and report $2\{1 - F_{t_{n-1}}(|T_{\text{obs}}|)\}$. The test is exactly level α under the normal model; outside normality it is generally an approximation.

C. Null-centered bootstrap test for a mean

Hypotheses. $H_0 : \mu = \mu_0$, usually against a two-sided alternative.

Null construction. Center the empirical observations at the null:

$$X_i^0 = X_i - \bar{X} + \mu_0.$$

This empirical distribution has mean μ_0 . Draw n values with replacement from $\{X_i^0\}$ and compute a studentized statistic for each resample:

$$T_b^* = \frac{\bar{X}_b^* - \mu_0}{S_b^*/\sqrt{n}}, \quad b = 1, \dots, B.$$

The observed statistic is $T_{\text{obs}} = (\bar{X} - \mu_0)/(S/\sqrt{n})$.

Monte Carlo p-value. For a symmetric two-sided comparison,

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{|T_b^*| \geq |T_{\text{obs}}|\}}{B + 1}.$$

For an upper-tail test replace the indicator by $\mathbf{1}\{T_b^* \geq T_{\text{obs}}\}$; for a lower-tail test use $\mathbf{1}\{T_b^* \leq T_{\text{obs}}\}$. The +1 correction prevents a zero Monte Carlo p-value. It does not make this bootstrap test finite-sample exact. Report B and, when relevant, Monte Carlo uncertainty.

Assumptions and interpretation. Independent sampling and regularity conditions for the statistic are needed; finite moments and a sample large enough to represent the population tails

matter. Resampling the raw data without centering does not impose H_0 and does not produce a null test distribution. Bootstrap calibration estimates uncertainty; it cannot fix confounding, selection bias, or dependence.

2.3 One-sample variance

A. Exact chi-square test under normality

Model and hypotheses. Assume $X_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$. Test $H_0 : \sigma^2 = \sigma_0^2$ against $\sigma^2 > \sigma_0^2$, $\sigma^2 < \sigma_0^2$, or $\sigma^2 \neq \sigma_0^2$.

Statistic and null law.

$$Q = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2 \quad \text{under } H_0.$$

Procedure. For $H_1 : \sigma^2 > \sigma_0^2$, reject when $Q > \chi_{n-1, 1-\alpha}^2$; p-value $1 - F_{\chi_{n-1}^2}(Q_{\text{obs}})$. For a lower-tail alternative, reject when $Q < \chi_{n-1, \alpha}^2$. For a two-sided alternative, reject if $Q < \chi_{n-1, \alpha/2}^2$ or $Q > \chi_{n-1, 1-\alpha/2}^2$. The equal-tail p-value is $\min\{1, 2 \min(F(Q_{\text{obs}}), 1 - F(Q_{\text{obs}}))\}$.

Assumptions. This is exact only for iid normal data. The chi-square calibration is particularly sensitive to nonnormality; a large sample alone does not make this finite-sample identity exact.

With an unrestricted population shape, there is no general finite-sample, distribution-free test whose null is only $\sigma^2 = \sigma_0^2$: distributions with the same variance can have very different shapes. A variance test therefore needs a parametric shape assumption, an asymptotic moment condition, or a resampling approximation with an explicit shape-preservation assumption.

B. Asymptotic test for a variance

Assume $X_1, \dots, X_n$ are iid with mean μ , variance $\sigma^2 > 0$, and finite fourth central moment $\mu_4 = \mathbb{E}[(X - \mu)^4]$. We derive the limiting distribution of the sample variance rather than treating it as a standard formula.

Write $Y_i = X_i - \mu$ and define $\bar{Y} = n^{-1} \sum_i Y_i$ and $B_n = n^{-1} \sum_i Y_i^2$. Since $\sum_i (X_i - \bar{X})^2 = \sum_i (Y_i - \bar{Y})^2$,

$$S^2 = \frac{1}{n-1} \sum_i (Y_i - \bar{Y})^2 = \frac{n}{n-1} (B_n - \bar{Y}^2).$$

Therefore,

$$\sqrt{n}(S^2 - \sigma^2) = \sqrt{n}(B_n - \sigma^2) - \sqrt{n}\bar{Y}^2 + o_p(1).$$

To see why the last two simplifications are valid, $B_n = O_p(1)$, so multiplication by $n/(n-1)$ changes it by $O_p(n^{-1})$, which becomes $o_p(n^{-1/2})$ on the $\sqrt{n}$ scale. Also $\bar{Y} = O_p(n^{-1/2})$, so $\sqrt{n}\bar{Y}^2 = O_p(n^{-1/2}) = o_p(1)$. The leading random term is thus the centered average of Y_i^2 .

The ordinary CLT applied to Y_i^2 , which has mean σ^2 and variance

$$\text{Var}(Y_i^2) = \mathbb{E}[Y_i^4] - \{\mathbb{E}[Y_i^2]\}^2 = \mu_4 - \sigma^4,$$

gives

$$\sqrt{n}(S^2 - \sigma^2) \Rightarrow N(0, \tau^2), \quad \tau^2 = \mu_4 - \sigma^4.$$

Here σ^4 means $(\sigma^2)^2$, not a fourth moment.

Estimate the asymptotic variance. Define the sample fourth central moment

$$m_4 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^4.$$

By the law of large numbers, $m_4 \rightarrow_p \mu_4$ and $S^2 \rightarrow_p \sigma^2$. Hence

$$\hat{\tau}^2 = m_4 - (S^2)^2$$

is a consistent estimator of $\tau^2 = \mu_4 - \sigma^4$. Set $\hat{\tau} = \sqrt{\hat{\tau}^2}$ when this estimate is positive. No finite-sample correction is needed for consistency or for this asymptotic test. The phrase “finite-sample correction” is omitted because it would require naming a particular estimator and does not turn the asymptotic test into an exact one. In a small sample, $\hat{\tau}^2$ can be nonpositive from sampling variability; then this plug-in standard error is unusable, which is a warning that the asymptotic approximation/estimation is unstable, not a reason to silently alter the formula.

Test procedure. To test $H_0 : \sigma^2 = \sigma_0^2$, calculate

$$Z_V = \frac{\sqrt{n}(S^2 - \sigma_0^2)}{\hat{\tau}}.$$

Under H_0 and the stated moment conditions, $Z_V \Rightarrow N(0, 1)$. For an upper-tail alternative reject if $Z_V > z_{1-\alpha}$ and report $1 - \Phi(Z_V)$; for a lower-tail alternative reject if $Z_V < z_\alpha$ and report $\Phi(Z_V)$; for a two-sided alternative reject if $|Z_V| > z_{1-\alpha/2}$ and report $2\{1 - \Phi(|Z_V|)\}$. The approximation depends on a finite fourth moment and a stable estimate of it. If $\mu_4 = \infty$, the stated CLT need not hold.

C. Null bootstrap for a variance

The null bootstrap must generate data with variance σ_0^2 . The following residual-rescaling procedure does that while retaining the observed residual shape. It gives an approximate test, not a finite-sample exact one.

1. **State the null and choose the tail.** Test $H_0 : \sigma^2 = \sigma_0^2$ against $\sigma^2 > \sigma_0^2$, $\sigma^2 < \sigma_0^2$, or $\sigma^2 \neq \sigma_0^2$. Choose a significance level α and a large number of replicates B .
2. **Compute the observed statistic.** From the data calculate S^2 and set $V_{obs} = S^2 - \sigma_0^2$ (equivalently, use $\sqrt{n}V_{obs}$; the factor $\sqrt{n}$ does not change tail ranks). Positive values support the upper alternative; negative values support the lower alternative.
3. **Center residuals.** Calculate $e_i = X_i - \bar{X}$. These residuals have sample mean zero, so location does not enter the scale null.
4. **Impose the null variance.** Let $Z_i = (\sigma_0/S)e_i$. Since $S^2 = (n-1)^{-1} \sum_i e_i^2$, the empirical distribution on $Z_1, \dots, Z_n$ has sample variance exactly σ_0^2 using the same denominator $n-1$. It also keeps the empirical residual shape, up to a scale change.
5. **Generate null bootstrap samples.** For $b = 1, \dots, B$, sample n values with replacement from $\{Z_1, \dots, Z_n\}$, calling them $Z_{b1}^*, \dots, Z_{bn}^*$. Compute their sample variance S_b^{*2} and set $V_b^* = S_b^{*2} - \sigma_0^2$.

6. **Calculate the Monte Carlo p-value.** For an upper-tail alternative use

$$\hat{p}_U = \frac{1 + \sum_{b=1}^B \mathbf{1}\{V_b^* \geq V_{obs}\}}{B + 1},$$

for a lower-tail alternative use

$$\hat{p}_L = \frac{1 + \sum_{b=1}^B \mathbf{1}\{V_b^* \leq V_{obs}\}}{B + 1}.$$

For a two-sided test, one explicit equal-tail convention is $\hat{p}_2 = \min\{1, 2 \min(\hat{p}_L, \hat{p}_U)\}$. This remains a Monte Carlo approximation to a bootstrap test; the +1 avoids a zero reported p-value.

7. **Decide and report.** Reject H_0 when the chosen p-value is at most α . Report S^2 , σ_0^2 , the alternative, B , the p-value convention, and the resampling unit (one independent patient per draw).

Assumptions and limits. The procedure treats the centered empirical residual shape, after rescaling, as a useful estimate of the unknown shape under the null. It is approximate and needs enough independent observations to represent the tails. It is not a distribution-free exact test: the null specifies only a variance, and many different shapes can share that variance. If normality is credible, the chi-square variance test is simpler and exact. If the fourth moment is finite and the sample is large, the preceding asymptotic test is another option.

2.4 One-sample proportion

Let $R_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$, $K = \sum_i R_i$, and $\hat{p} = K/n$.

A. Exact binomial test

Test $H_0 : p = p_0$ against $p > p_0$, $p < p_0$, or $p \neq p_0$. Under H_0 , $K \sim \text{Bin}(n, p_0)$. For the upper-tail alternative, the p-value is $\sum_{k=K_{obs}}^n \binom{n}{k} p_0^k (1 - p_0)^{n-k}$; reject if this is at most α . For the lower tail sum from 0 through K_{obs} . For a two-sided test, one convention sums probabilities of outcomes whose null probability is no greater than the observed outcome's probability; software implementations may use this probability-ordering convention. Because K is discrete, the test is generally conservative: its size is at most, not necessarily equal to, α .

B. Score test / normal approximation

For $H_0 : p = p_0$, use

$$Z_P = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

Under H_0 and adequate expected success/failure counts, $Z_P \approx N(0, 1)$. Use the same normal-tail rejection rules as for a mean z test. The Wald statistic replaces p_0 in the standard error by $\hat{p}$; it can have poor size and coverage near 0 or 1, so the score or exact test is preferable for a point null.

C. Parametric null bootstrap

Under $H_0 : p = p_0$, draw $K_b^* \sim \text{Bin}(n, p_0)$ and calculate $T_b^* = (K_b^*/n - p_0)/\sqrt{p_0(1 - p_0)/n}$. Compare the simulated statistics to the observed score statistic using the tail specified by H_1 . This simulates the exact null law, so it is a Monte Carlo approximation to the exact binomial test; it does not improve on evaluating the binomial tails directly.

2.5 One-sample nonparametric location tests

These tests are useful when a mean model is implausible or the target is a median/location rather than a mean. They do not automatically test the same null as a t test.

A. Sign test for a median

Hypotheses. $H_0 : m = m_0$ versus a larger, smaller, or different population median m .

Statistic. Remove ties $X_i = m_0$ and let n_* be the number remaining. Define

$$K_+ = \sum_{i: X_i \neq m_0} \mathbf{1}\{X_i > m_0\}.$$

Null law and procedure. If observations are independent and the population is continuous with median m_0 , then $K_+ \sim \text{Bin}(n_*, 1/2)$. For $m > m_0$, use the upper binomial tail; for $m < m_0$, use the lower tail. One two-sided convention is $p = \min\{1, 2 \min[\mathbb{P}(B \leq K_{+,obs}), \mathbb{P}(B \geq K_{+,obs})]\}$ for $B \sim \text{Bin}(n_*, 1/2)$. Reject if the p-value is at most α . Because the distribution is discrete, the test may be conservative.

What it tests. Under continuity, this tests whether half the population lies on either side of m_0 , hence the median. It uses only signs and requires neither normality nor symmetry. It does not test the population mean. Ties require an explicit convention; discarding them is standard when they are sampling ties at the hypothesized median.

B. Wilcoxon signed-rank test

Hypotheses and assumption. For a one-sample location model, test whether the distribution is symmetric about m_0 (equivalently, zero location shift under a symmetric location family). Let $D_i = X_i - m_0$. Assume independent differences from a continuous distribution symmetric about zero.

Statistic. Discard zero D_i under the chosen convention. Rank $|D_i|$ from 1 to n_* , assigning average ranks to ties, and form

$$W^+ = \sum_{i=1}^{n_*} R_i \mathbf{1}\{D_i > 0\}, \quad W^- = \sum_i R_i - W^+.$$

Large W^+ supports a positive shift, small W^+ a negative shift; a two-sided statistic can be $\min(W^+, W^-)$ or the centered absolute deviation $|W^+ - \mathbb{E}_0 W^+|$.

Exact procedure. Under symmetry and continuity, conditional on absolute magnitudes, each of the 2^{n_*} sign patterns is equally likely. For a two-sided test, use the statistic $C = |W^+ - n_*(n_* + 1)/4|$.

Enumerate all $s = (s_1, \dots, s_{n_*}) \in \{-1, 1\}^{n_*}$, compute $W_s^+ = \sum_i R_i \mathbf{1}\{s_i = 1\}$ and $C_s = |W_s^+ - n_*(n_* + 1)/4|$, then report $2^{-n_*} \sum_s \mathbf{1}\{C_s \geq C_{obs}\}$. For a one-sided test, use the corresponding upper or lower tail of W^+ . This is an exact (possibly conservative because discrete) signed-rank p-value for the chosen extremeness rule. Ties and zero handling affect the reference distribution and must be specified.

Large-sample procedure. With no ties or zeros,

$$\mathbb{E}_0(W^+) = \frac{n_*(n_* + 1)}{4}, \quad \text{Var}_0(W^+) = \frac{n_*(n_* + 1)(2n_* + 1)}{24}.$$

Standardize W^+ by these moments and use a normal reference, with a continuity correction if desired. With ties, compute the null variance from the tied ranks or use permutation/sign enumeration.

Interpretation. It uses magnitudes through ranks and is often more powerful than the sign test under symmetry. Without symmetry it is not, in general, a test of the median or mean alone. A symmetric distribution with finite mean has mean and median at its symmetry center; that is why location interpretations can coincide under the assumption.

C. One-sample sign-flip test for a mean/location

Why use it? The signed-rank test sign-flips the *ranks* of the magnitudes. A sign-flip test for a mean instead sign-flips the original magnitudes and uses a mean-type statistic. It is therefore a distinct procedure and should not be conflated with either the sign test or Wilcoxon signed-rank test.

Hypotheses and model-based assumption. Let $D_i = X_i - \mu_0$. The model-based exact null is that the distribution of D_i is symmetric about zero, with the D_i independent. If $\mathbb{E}|D_i| < \infty$, this implies $\mathbb{E}[D_i] = 0$; the converse is false. Thus this is not an exact test of the mean-zero null alone.

Statistic. A natural choice is the sample mean $T_{obs} = \bar{D} = n^{-1} \sum_i D_i$. A studentized alternative is $T_{obs} = \bar{D}/(S_D/\sqrt{n})$. The choice should be fixed before looking at the results.

Exact sign-flip procedure. Conditional on the magnitudes $|D_1|, \dots, |D_n|$, symmetry implies that the signs are independent fair signs. For each sign vector $s = (s_1, \dots, s_n) \in \{-1, +1\}^n$, calculate

$$T_s = \frac{1}{n} \sum_{i=1}^n s_i |D_i|.$$

There are 2^n equally likely sign vectors. For $H_1 : \mathbb{E}[D_i] > 0$, the exact p-value is $2^{-n} \sum_s \mathbf{1}\{T_s \geq T_{obs}\}$; for $H_1 : \mathbb{E}[D_i] < 0$, use $2^{-n} \sum_s \mathbf{1}\{T_s \leq T_{obs}\}$. For a two-sided alternative, use the explicit extremeness rule $p = 2^{-n} \sum_s \mathbf{1}\{|T_s| \geq |T_{obs}|\}$. Reject when the selected p-value is at most α . Discreteness can make the test conservative. If n is too large to enumerate, sample B independent sign vectors and use the corresponding Monte Carlo tail fraction with the +1 correction.

Randomization justification. A sign-flip test is also exact for a design-based sharp null when the signs arise from actual random assignment. For example, in a paired crossover trial with treatment order randomized independently within pairs, swapping the order in each pair changes D_i to $-D_i$. Enumerating the allowed within-pair swaps gives the randomization distribution under the sharp null that treatment has no effect for any patient. This design-based exactness does not require a symmetric population distribution; it requires the stated assignment mechanism and sharp null.

Scope. In a one-arm study without randomized signs, the model-based test is justified by symmetry, not by random sampling alone. A skewed distribution can have mean zero but is not invariant

under sign changes, so sign-flipping need not control type-I error for the weak null $\mathbb{E}[D_i] = 0$. If the scientific target is only the mean and symmetry is not credible, use a t/asymptotic test or a null-centered bootstrap designed for that mean null.

2.6 High-dimensional one-sample inference

Suppose each patient has a vector $\mathbf{X}_i \in \mathbb{R}^p$ of biomarkers or clinical outcomes with mean vector $\boldsymbol{\mu}$. A global hypothesis is $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$.

A. One-sample Hotelling T^2

Assume iid multivariate normal observations with nonsingular covariance Σ and $n > p$. Define

$$T^2 = n(\bar{\mathbf{X}} - \boldsymbol{\mu}_0)^\top S^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}_0).$$

Under H_0 , $\frac{n-p}{p(n-1)}T^2 \sim F_{p,n-p;1-\alpha}$. Reject for values exceeding $F_{p,n-p;1-\alpha}$. This is an exact global test under multivariate normality. It says the mean vector differs somewhere; it does not identify which endpoint differs. If $p \geq n$, S^{-1} does not exist.

B. Coordinate-wise and max-statistic procedures

For each endpoint j , compute $T_j = \sqrt{n}(\bar{X}_j - \mu_{0j})/S_j$ and its one-dimensional p-value. To control family-wise error across p tests, Bonferroni rejects endpoint j if $p_j \leq \alpha/p$; Holm orders p-values $p_{(1)} \leq \dots \leq p_{(p)}$ and rejects sequentially while $p_{(k)} \leq \alpha/(p - k + 1)$. Benjamini–Hochberg targets false discovery rate: order p-values and reject through the largest k satisfying $p_{(k)} \leq kq/p$ (under independence or standard positive-dependence conditions).

To account for endpoint dependence, use a max statistic $M = \max_j |T_j|$. A null bootstrap resamples whole centered patient vectors $\mathbf{X}_i - \bar{\mathbf{X}} + \boldsymbol{\mu}_0$, preserving within-patient endpoint correlations, and recomputes M_b^* . The adjusted p-value for endpoint j is the fraction of bootstrap maxima at least $|T_{j,obs}|$, with the +1 correction. This gives simultaneous, dependence-aware inference approximately, subject to bootstrap regularity and a suitable sample size.

3 Session 2: two-sample tests

3.1 Independent, paired, and randomized designs

For independent treatment and control samples, let $X_1, \dots, X_m$ be outcomes in treatment and $Y_1, \dots, Y_n$ in control; assume independence across units and arms. The mean contrast is $\Delta = \mu_X - \mu_Y$. A randomized trial supports a causal treatment interpretation under the randomization, consistency, no-interference, and appropriate missing-data assumptions. In observational samples, equality tests describe association unless additional causal assumptions are justified.

For paired or crossover data, patient i contributes a pair (X_i, Y_i) . Define $D_i = X_i - Y_i$. Inference for the paired contrast is one-sample inference on the independent patient-level differences $D_1, \dots, D_N$. Do not treat paired observations as independent arms.

3.2 Two independent means

A. Welch test: unequal variances

Hypotheses. $H_0 : \mu_X - \mu_Y = \Delta_0$ against greater, smaller, or unequal contrast.

Statistic.

$$T_W = \frac{(\bar{X} - \bar{Y}) - \Delta_0}{\sqrt{S_X^2/m + S_Y^2/n}}, \quad \nu = \frac{(S_X^2/m + S_Y^2/n)^2}{(S_X^2/m)^2/(m-1) + (S_Y^2/n)^2/(n-1)}.$$

Procedure. Compare T_W with a t_ν reference distribution (Welch–Satterthwaite approximation). For a two-sided test, reject if $|T_W| > t_{\nu, 1-\alpha/2}$; p-value $2[1 - F_{t_\nu}(|T_{W,obs}|)]$. For one-sided tests use the corresponding tail. Under independent normal samples with unequal variances, this is generally approximate; with large independent samples it is asymptotically justified by the two-sample CLT.

Assumptions. Independent samples, finite group variances, and adequate sample sizes for a useful approximation. Welch does not require equal variances. Strong skewness with small group sizes can still undermine its calibration.

B. Pooled two-sample t test: common variance

Assume independent normal samples with $\text{Var}(X_i) = \text{Var}(Y_j) = \sigma^2$. Test $H_0 : \mu_X - \mu_Y = \Delta_0$ with

$$S_p^2 = \frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2}, \quad T_p = \frac{(\bar{X} - \bar{Y}) - \Delta_0}{S_p \sqrt{1/m + 1/n}}.$$

Under H_0 , $T_p \sim t_{m+n-2}$ exactly. Use the appropriate one- or two-sided t tail and critical value. If variances are unequal, especially with unequal sample sizes, this pooled calibration can have incorrect size. Do not select it after a preliminary variance test; use Welch unless common variance is scientifically justified.

C. Asymptotic z test

Under independent sampling with finite variances and both sample sizes growing,

$$Z_W = \frac{(\bar{X} - \bar{Y}) - \Delta_0}{\sqrt{S_X^2/m + S_Y^2/n}} \Rightarrow N(0, 1).$$

Use normal critical values and tails. This is the large-sample limit underlying Welch's procedure; the Welch t reference often gives a better finite-sample approximation.

D. Bootstrap test for a mean contrast

The null is $H_0 : \mu_X - \mu_Y = \Delta_0$. One null-imposing scheme centers each group:

$$X_i^0 = X_i - \bar{X} + c_X, \quad Y_j^0 = Y_j - \bar{Y} + c_Y, \quad c_X - c_Y = \Delta_0.$$

For example choose $c_Y = 0$, $c_X = \Delta_0$. Resample independently within the two centered groups, preserving their sample sizes, and calculate a studentized contrast T_b^* . The two-sided p-value is $(1 + \sum_b \mathbf{1}\{|T_b^*| \geq |T_{obs}|\})/(B + 1)$; use one-sided tails as appropriate.

This imposes equal means at the specified contrast while retaining the groups' empirical shapes and variances. Pooling both groups and resampling from a common distribution imposes the stronger null $F_X = F_Y$, not merely equal means. Bootstrap validity is approximate and depends on independent sampling and regularity conditions. For paired data, resample whole pairs or resample D_i ; never resample the two paired columns independently.

3.3 Two independent proportions

Let $K_X \sim \text{Bin}(m, p_X)$ and $K_Y \sim \text{Bin}(n, p_Y)$ independently, with $\hat{p}_X = K_X/m$ and $\hat{p}_Y = K_Y/n$.

A. Pooled score test

Test $H_0 : p_X = p_Y$ (equivalently $p_X - p_Y = 0$) against a one- or two-sided alternative. Under the null, estimate the common probability by $\hat{p}_0 = (K_X + K_Y)/(m + n)$. The score statistic is

$$Z_{2P} = \frac{\hat{p}_X - \hat{p}_Y}{\sqrt{\hat{p}_0(1 - \hat{p}_0)(1/m + 1/n)}}.$$

With adequate expected successes and failures, $Z_{2P} \approx N(0, 1)$. Use its normal tails. Pooling is correct for this equality-null test because the null imposes a common response probability. For a confidence interval on $p_X - p_Y$, use an unpooled standard error or a more accurate interval method; interval construction is a distinct problem.

B. Fisher exact test

Write the 2×2 table with treated/control rows and response/nonresponse columns. Under $H_0 : p_X = p_Y$, condition on the total number of responses $K = K_X + K_Y$. Given the margins, the treated response count K_X has a hypergeometric distribution:

$$\mathbb{P}(K_X = k \mid K) = \frac{\binom{m}{k} \binom{n}{K-k}}{\binom{m+n}{K}},$$

over feasible k . For a one-sided alternative sum the appropriate hypergeometric tail. A common two-sided definition sums probabilities of tables no more likely under the null than the observed table. Reject when the p-value is at most α . This is a conditional exact test and is useful for sparse counts; discreteness can make it conservative. The precise two-sided convention should be reported.

C. Bootstrap under the equality null

For a parametric null bootstrap, draw $K_X^* \sim \text{Bin}(m, \hat{p}_0)$ and $K_Y^* \sim \text{Bin}(n, \hat{p}_0)$ independently, recompute the score statistic, and compare its simulated tails with the observed statistic. This is a Monte Carlo approximation to the pooled score null law. A nonparametric bootstrap that resamples each arm independently estimates uncertainty around the two observed rates and does not impose equality unless explicitly centered or otherwise constrained.

3.4 Two-sample variance and spread

A. Exact F test for equal variances

Assume independent normal samples. Test $H_0 : \sigma_X^2 = \sigma_Y^2$ against a one- or two-sided alternative with

$$F = \frac{S_X^2}{S_Y^2} \sim F_{m-1, n-1} \quad \text{under } H_0.$$

For $\sigma_X^2 > \sigma_Y^2$, reject in the upper tail; for the reverse alternative, use the lower tail. For a two-sided alternative, allocate $\alpha/2$ to each tail (or report the corresponding equal-tail p-value). This is exact under normality and sensitive to nonnormal shapes.

B. Levene and Brown–Forsythe tests

For group $g = 1, 2$, choose center C_g as the group mean (Levene) or median (Brown–Forsythe), and form absolute deviations $Z_{ig} = |X_{ig} - C_g|$. Conduct a one-way ANOVA on these deviations. With $N = m + n$ and group means $\bar{Z}_g$, define

$$F_L = \frac{\sum_g n_g (\bar{Z}_g - \bar{Z})^2 / (2 - 1)}{\sum_g \sum_i (Z_{ig} - \bar{Z}_g)^2 / (N - 2)}.$$

The reference is approximately $F_{1, N-2}$; reject for a large value. The median-centered version is more robust to skewness. The procedure tests equality of average absolute deviations from the chosen centers; it is a robust indicator of spread differences, not an exact test of equality of variances under arbitrary distributions.

C. Fligner–Killeen test

For G independent groups, let C_g be the group median and $A_{ig} = |X_{ig} - C_g|$. Pool all N deviations and let R_{ig} be their ranks. Transform ranks to normal scores, for example

$$a_{ig} = \Phi^{-1} \left(\frac{1}{2} + \frac{R_{ig}}{2(N+1)} \right).$$

Let $\bar{a}_g$ and $\bar{a}$ be group and overall score means, and $s_a^2 = (N-1)^{-1} \sum_{ig} (a_{ig} - \bar{a})^2$. The statistic is

$$X_{FK}^2 = \frac{\sum_g n_g (\bar{a}_g - \bar{a})^2}{s_a^2}.$$

Under the null of equal scale across groups, it is approximately χ_{G-1}^2 ; reject for a large value. It is a rank-based, robust scale test. Its clean scale interpretation assumes comparable distributional shapes; it does not test every possible variance difference without assumptions.

3.5 Paired tests as one-sample tests

Let $D_i = X_i - Y_i$ for N independent pairs. To test mean difference δ_0 , use

$$T_D = \frac{\bar{D} - \delta_0}{S_D / \sqrt{N}}.$$

Under iid normal differences this is exactly t_{N-1} under $H_0 : \mathbb{E}[D] = \delta_0$; with finite variance it is asymptotically normal/student- t . For nonparametric paired inference, test median difference with the sign test, or use signed-rank under symmetric differences. A sign-flip randomization test is exact if treatment labels were randomized within pairs under the sharp no-effect null. The paired analysis accounts for covariance: $\text{Var}(X_i - Y_i) = \text{Var}(X_i) + \text{Var}(Y_i) - 2 \text{Cov}(X_i, Y_i)$.

3.6 Permutation and randomization tests

A. Independent-sample permutation test

Null. $H_0 : F_X = F_Y$, equality of the entire outcome distributions. Under this null, conditional on the pooled outcomes, group labels are exchangeable.

Statistic. Choose a statistic aligned with the target, for example $T = \bar{X} - \bar{Y}$, a studentized difference, rank-sum W_X , or a CDF distance $D_{m,n}$. Keep group sizes fixed.

Procedure. Pool $N = m + n$ observations. Enumerate all $\binom{N}{m}$ assignments of m labels to treatment, compute T^π for each, and calculate the proportion at least as extreme as T_{obs} under the alternative's tail rule. For a two-sided difference-in-means test under $F_X = F_Y$, use $|\bar{X}^\pi - \bar{Y}^\pi| \geq |\bar{X} - \bar{Y}|$. This is exact conditional on the pooled values under exchangeability. The pooled-label procedure is not an exact test of a general nonzero mean contrast Δ_0 ; a nonzero contrast needs a location-shift model and an explicit null-imposing transformation. If enumeration is too large, sample B random assignments and use $(1 + \#\{T_b^* \text{ at least as extreme as } T_{obs}\})/(B + 1)$.

Important distinction. The raw permutation test is not generally exact for the weaker null $\mu_X = \mu_Y$ when variances or shapes differ. A studentized permutation statistic can be asymptotically valid for a weak mean null under suitable conditions. State whether the null is equal distributions or equal means.

B. Randomization test for a clinical trial

Suppose the trial randomized exactly m of N patients to treatment. Under the sharp null $H_0 : Y_i(1) = Y_i(0)$ for every patient, each patient's outcome is fixed regardless of assignment. Reassign the treatment labels according to the actual design, compute a statistic such as the mean difference for each permitted assignment, and compare the observed statistic with this randomization distribution. The exhaustive randomization p-value is the fraction of allowed assignments at least as extreme as observed. It is exact under the sharp null and the stated randomization scheme. Under a weak null of zero average effect while individual effects vary, this test is not generally exact; studentized randomization methods can offer asymptotic validity.

C. Paired sign-flip test

For paired differences D_i , use $T = \bar{D}$ or a studentized mean difference. If each pair's treatment order was randomized, enumerate all 2^N within-pair label swaps, equivalent to replacing each D_i by $s_i D_i$ for $s_i \in \{-1, 1\}$. The exact test is for the sharp null of no effect for any patient. As a model-based test for a population contrast, sign flips require the null distribution of differences to be symmetric about zero; mean zero alone does not suffice.

3.7 Two-sample nonparametric tests

The procedures below are called nonparametric because they do not specify a normal or other named parametric family for the outcome distribution. They are not assumption-free. First decide whether the scientific target is equality of full distributions, a rank-based probability of superiority, a location shift, a median, or a paired change in mean. The tests below have different null hypotheses and are not interchangeable.

A. Wilcoxon rank-sum / Mann–Whitney test

Design and null hypothesis. There are two independent samples, $X_1, \dots, X_m \stackrel{\text{iid}}{\sim} F_X$ and $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} F_Y$, with $N = m + n$. The exact permutation null is

$$H_0 : F_X = F_Y \quad \text{against} \quad H_1 : F_X \neq F_Y.$$

Under this null, the group labels are exchangeable conditional on the pooled observations. Equality of means or equality of medians alone is a weaker null and does not in general justify the exact pooled-label permutation distribution.

Statistic and its connection to pairwise comparisons. Pool the observations and assign ranks $R_1, \dots, R_N$, using average ranks for ties. Let W_X be the sum of the ranks attached to the X observations. Define

$$U_X = \sum_{i=1}^m \sum_{j=1}^n \left[\mathbf{1}\{X_i > Y_j\} + \frac{1}{2} \mathbf{1}\{X_i = Y_j\} \right].$$

Each X – Y pair contributes one to U_X if $X_i > Y_j$, zero if $X_i < Y_j$, and one half for a tie. The rank-sum identity is

$$W_X = \frac{m(m+1)}{2} + U_X.$$

To derive it, write each X_i 's midrank as one plus the number of pooled observations below it plus half the number tied with it (excluding itself). Summing the within- X contributions gives $m(m-1)/2$ regardless of their ordering; the cross-group contributions give U_X . Adding the m ones yields the identity. Thus $U_X/(mn)$ estimates the probability-of-superiority quantity $\mathbb{P}(X > Y) + \frac{1}{2}\mathbb{P}(X = Y)$.

Exact conditional test: complete procedure.

1. Pool the observed values, retain the group sizes m and n , and calculate the observed rank sum $W_{X,obs}$ (or $U_{X,obs}$).
2. Under $H_0 : F_X = F_Y$, allocate the N pooled observations to an X group of size m in every possible way. There are $\binom{N}{m}$ allocations. With ties, keep the observed pooled values and their tied midranks fixed; only the labels are reassigned.
3. For each allocation π , compute its rank sum W_X^π . For an upper-tail alternative use $W_X^\pi \geq W_{X,obs}$; for a lower-tail alternative use $W_X^\pi \leq W_{X,obs}$. For the two-sided convention used here, calculate

$$C_{obs} = \left| W_{X,obs} - \frac{m(N+1)}{2} \right|, \quad p_{2s} = \frac{\#\{\pi : |W_X^\pi - m(N+1)/2| \geq C_{obs}\}}{\binom{N}{m}}.$$

This explicitly names the two-sided extremeness rule; other exact two-sided conventions can give different p-values for discrete distributions.

4. Reject the stated null at level α if the p-value is at most α . This is an exact conditional test because, under $F_X = F_Y$, each allocation of the labels is equally probable given the pooled values.

If $\binom{N}{m}$ is too large, sample B allocations uniformly. Let K be the number of sampled allocations at least as extreme as observed. Estimate the tail probability by

$$\hat{p}_{MC} = \frac{1 + K}{B + 1}.$$

The +1 convention avoids a zero Monte Carlo p-value; it does not make an incomplete Monte Carlo enumeration exact.

Derivation of the large-sample normal calibration. Under the permutation null, group X receives a simple random sample of m ranks without replacement from $1, \dots, N$. The mean rank is $(N + 1)/2$. Therefore,

$$\mathbb{E}_0(W_X) = m \frac{N + 1}{2}.$$

The finite-population variance of the ranks $1, \dots, N$ is $(N^2 - 1)/12$. The variance of their sum in a size- m sample without replacement is

$$\text{Var}_0(W_X) = \frac{mn}{N - 1} \frac{N^2 - 1}{12} = \frac{mn(N + 1)}{12}.$$

Consequently, as both sample sizes grow,

$$Z_W = \frac{W_X - m(N + 1)/2}{\sqrt{mn(N + 1)/12}} \sim N(0, 1).$$

For tied pooled values with tie-block sizes $t_1, \dots, t_L$, the pooled midranks are still fixed under label permutations. The conditional random-sampling variance is

$$\text{Var}_0(W_X) = \frac{mn}{N(N - 1)} \sum_{k=1}^N \left(R_k - \frac{N + 1}{2} \right)^2 = \frac{mn}{12} \left[N + 1 - \frac{\sum_{\ell=1}^L (t_\ell^3 - t_\ell)}{N(N - 1)} \right].$$

Use this tie-corrected variance in Z_W . With untied integer ranks, a continuity correction subtracts $1/2$ from the absolute observed deviation before dividing by its standard deviation for a two-sided test (or shifts by $1/2$ toward the null for a one-sided tail). With tied midranks, the support can have half-unit spacing, so use the appropriate lattice correction or omit it. Calculate the normal-tail p-value from this standardized statistic and reject when it is at most α . The normal approximation is not an exact finite-sample law.

Interpretation and limits. Under a common-shape location-shift model, this procedure can be interpreted as testing a zero shift. Without that model, it is a test of the equality-of-distributions null, and $U_X/(mn)$ estimates a probabilistic superiority contrast. It is not generally a test of equal means or equal medians. A pooled permutation calibration is not finite-sample exact under merely equal means when the group distributions differ in shape or variance.

B. Two-sample Kolmogorov–Smirnov test

Design and null hypothesis. The samples are independent and the null is equality of the entire cumulative distribution functions:

$$H_0 : F_X(t) = F_Y(t) \text{ for every } t \quad \text{against} \quad H_1 : F_X \neq F_Y.$$

This is stronger than equality of means or medians. The two-sided KS test can detect a difference in location, scale, or shape.

Statistic. Define the empirical CDFs

$$\hat{F}_X(t) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{X_i \leq t\}, \quad \hat{F}_Y(t) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{Y_j \leq t\}.$$

The two-sided statistic is the largest vertical separation between the two empirical CDFs:

$$D_{m,n} = \sup_t |\hat{F}_X(t) - \hat{F}_Y(t)|.$$

For a directional alternative, use $D^+ = \sup_t (\hat{F}_X - \hat{F}_Y)$ or $D^- = \sup_t (\hat{F}_Y - \hat{F}_X)$, stating the direction in advance.

Exact conditional/permutation derivation and procedure. Under H_0 and continuous outcomes, order the pooled observations as $Z_{(1)} < \dots < Z_{(N)}$. Conditional on these values, the sequence of X and Y labels is uniformly distributed over the $\binom{N}{m}$ sequences containing m labels X . Let A_k be the number of X labels among the first k pooled observations. Immediately after $Z_{(k)}$,

$$\hat{F}_X(Z_{(k)}) - \hat{F}_Y(Z_{(k)}) = \frac{A_k}{m} - \frac{k - A_k}{n} = \frac{NA_k - mk}{mn}.$$

Between consecutive pooled observations both empirical CDFs are constant, so the supremum occurs at a pooled observation (including the equivalent left limits). Thus the conditional null distribution of $D_{m,n}$ is obtained by enumerating the label sequences and calculating

$$D^\pi = \max_{0 \leq k \leq N} \left| \frac{NA_k^\pi - mk}{mn} \right|.$$

The exact two-sided p-value is $\#\{\pi : D^\pi \geq D_{obs}\} / \binom{N}{m}$. This yields the following procedure:

1. Compute both empirical CDFs and D_{obs} .
2. Pool the outcomes and enumerate every allocation of m labels to the X sample, keeping m, n fixed.
3. Recompute D^π for each allocation and count allocations with $D^\pi \geq D_{obs}$.
4. Divide by $\binom{N}{m}$ to obtain the exact conditional p-value; reject H_0 when it is at most α .

For large $\binom{N}{m}$, randomly sample allocations and use the +1 Monte Carlo p-value correction described above. With ties/discrete outcomes, the continuous-theory KS calibration below is not exact. A pooled-label permutation test remains valid under exchangeability when $F_X = F_Y$, with ties retained in every permutation.

Derivation of the asymptotic null distribution. Under H_0 and continuity, the two empirical processes converge jointly to Brownian bridges. Their scaled difference converges to a standard Brownian bridge $B(t)$:

$$\sqrt{\frac{mn}{m+n}} \{\hat{F}_X(t) - \hat{F}_Y(t)\} \Rightarrow B(F(t)).$$

Since F maps the support onto $[0, 1]$, the limiting statistic is $K = \sup_{0 \leq u \leq 1} |B(u)|$. Its CDF is

$$\mathbb{P}(K \leq x) = 1 - 2 \sum_{r=1}^{\infty} (-1)^{r-1} e^{-2r^2 x^2}, \quad x > 0.$$

Thus an asymptotic p-value is $\mathbb{P}(K \geq \sqrt{mn/(m+n)} D_{obs})$. Reject for a large scaled KS statistic (equivalently, a small tail p-value). This limit assumes independent samples from continuous distributions under the equality null; use the permutation calibration for finite samples or ties.

Interpretation. KS reports the largest CDF gap but does not identify where or why the distributions differ. It is an omnibus distributional test, not a test of the mean alone.

C. Paired sign and Wilcoxon signed-rank tests

For paired measurements, the independent units are patients, not the two columns. Let $D_i = X_i - Y_i$, $i = 1, \dots, N_p$, be the within-patient differences. These tests are one-sample tests on the differences and should not be confused with the independent-sample rank-sum test above.

Sign test. The target is the median difference. Under a continuous distribution with median zero, $\mathbb{P}(D_i > 0) = \mathbb{P}(D_i < 0) = 1/2$. Drop zero differences and let $N_* = \sum_i \mathbf{1}\{D_i \neq 0\}$ and $K_+ = \sum_i \mathbf{1}\{D_i > 0\}$. Independence of patients implies

$$K_+ \mid N_* \sim \text{Bin}(N_*, 1/2) \quad \text{under } H_0.$$

For H_1 : median difference > 0 , calculate $\sum_{k=K_{+,obs}}^{N_*} \binom{N_*}{k} 2^{-N_*}$; for a lower-tail alternative sum from 0 to $K_{+,obs}$. For a two-sided alternative, one equal-tail convention is twice the smaller binomial tail, capped at 1. Reject if the selected p-value is at most α . This procedure uses only the directions of differences: it does not test a zero mean and can be less powerful than signed-rank when magnitudes are informative.

Wilcoxon signed-rank test. This uses both signs and magnitudes and requires the distribution of D_i to be symmetric about the null location 0 (or about a hypothesized δ_0 after replacing D_i by $D_i - \delta_0$). Under this symmetry null, conditional on the absolute differences, each sign is an independent fair coin. This conditional sign argument gives the exact null law.

1. Remove zero differences and set N_* to the number remaining. Rank $|D_i|$ from smallest to largest; use average ranks for tied absolute values. Call the rank r_i .
2. Compute the positive-rank statistic $W^+ = \sum_{i:D_i > 0} r_i$. For a directional alternative, large W^+ supports positive shifts and small W^+ supports negative shifts.
3. Under the symmetry null, condition on the observed absolute differences (and hence the ranks). Each of the 2^{N_*} sign assignments is equally likely. Enumerate them, calculate W^{+*} for each, and for a two-sided test compare $|W^{+*} - \frac{1}{2} \sum_i r_i|$ with $|W_{obs}^+ - \frac{1}{2} \sum_i r_i|$. The p-value is the proportion at least as extreme; use the corresponding upper/lower tail for a one-sided alternative.
4. Reject if the selected p-value is at most α . The exact test is conditional on the magnitudes and is model-based through symmetry; it is not exact merely because the mean difference is zero.

With no tied absolute differences, the ranks are $1, \dots, N_*$. Since $W^+ = \sum_i r_i B_i$ with independent $B_i \sim \text{Bernoulli}(1/2)$ under the null,

$$\mathbb{E}_0(W^+) = \frac{1}{2} \sum_{i=1}^{N_*} r_i = \frac{N_*(N_* + 1)}{4}, \quad \text{Var}_0(W^+) = \frac{1}{4} \sum_{i=1}^{N_*} r_i^2 = \frac{N_*(N_* + 1)(2N_* + 1)}{24}.$$

With tied absolute differences, use midranks in the conditional enumeration; the mean remains $\frac{1}{2} \sum_i r_i$ and variance is $\frac{1}{4} \sum_i r_i^2$. For large N_* , standardize with this mean and variance and use a normal approximation, with a continuity correction if desired. The signed-rank test is often described as a test of a zero location shift; without symmetry, that interpretation is not justified, and it is not generally a test of the mean.

Design-based paired sign flips are a separate justification. If treatment order was randomized within each pair, swapping the two treatment labels flips the sign of D_i . Under the sharp null that neither treatment affects any patient, all 2^{N_p} sign assignments are equally likely by the design, so the same enumeration yields an exact randomization test for a chosen statistic. This design-based exactness comes from randomization, not from a symmetric outcome distribution. If treatment order was not randomized, sign flipping is not a design-based exact test; a model-based sign-flip test instead requires symmetric differences under its null.

3.8 High-dimensional two-sample inference

Suppose each patient has p endpoints, with vectors $\mathbf{X}_i$ in treatment and $\mathbf{Y}_j$ in control. The global mean-vector null is $H_0 : \boldsymbol{\mu}_X = \boldsymbol{\mu}_Y$.

A. Two-sample Hotelling T^2

Assume independent multivariate normal samples with a common nonsingular covariance matrix and $N = m + n > p + 1$. Define the pooled covariance S_p and

$$T^2 = \frac{mn}{N} (\bar{\mathbf{X}} - \bar{\mathbf{Y}})^\top S_p^{-1} (\bar{\mathbf{X}} - \bar{\mathbf{Y}}).$$

Under H_0 ,

$$\frac{N - p - 1}{p(N - 2)} T^2 \sim F_{p, N - p - 1}.$$

Reject for a large value. This is an exact global test under the multivariate normal/common-covariance model. If p is too large relative to N , the pooled covariance is singular or unstable, so the classical statistic is unavailable or unreliable.

B. Max- T and multiplicity-adjusted testing

For each endpoint compute a two-sample studentized statistic T_j . The family-wise max statistic is $M = \max_{1 \leq j \leq p} |T_j|$. In a randomized trial, permute treatment assignments according to the design, keeping each patient's entire vector together, and recompute all T_j and M for every assignment. The adjusted p-value for endpoint j is the proportion of assignments with $M^\pi \geq |T_{j, \text{obs}}|$. This controls family-wise error under the sharp randomization null. A bootstrap or multiplier approximation

can replace exhaustive randomization when justified; resample whole patient vectors to preserve endpoint correlations.

For unadjusted endpoint p-values, Bonferroni uses threshold α/p ; Holm uses sequential thresholds $\alpha/(p-k+1)$ for ordered p-values; Benjamini–Hochberg rejects through the largest k with $p_{(k)} \leq kq/p$, targeting false discovery rate under its dependence conditions. A global rejection does not itself identify an endpoint. Prespecify whether the trial seeks a global effect, sparse endpoint discoveries, or simultaneous endpoint-specific conclusions.

4 Base R procedures

The following code uses base R. Each test call still requires the assumptions and null stated in the notes. For randomization, keep a patient as the assignment/resampling unit and mirror the actual trial allocation scheme.

Listing 1: Classical one-sample and two-sample procedures

```
# One-sample mean: exact under normality; otherwise often asymptotic
t.test(change, mu = mu0, alternative = "two.sided")

# Exact one-sample binomial test
binom.test(responders, n_patients, p = p0,
            alternative = "greater")

# Exact normal-theory one-sample variance test
q <- (length(change) - 1) * var(change) / sigma0^2
p.upper <- pchisq(q, df = length(change) - 1,
                  lower.tail = FALSE)

# Independent means: Welch test (does not pool variances)
t.test(treatment, control, var.equal = FALSE)

# Paired mean comparison: one-sample t on within-patient differences
t.test(treatment, control, paired = TRUE)

# Independent proportions: pooled score test under equality
prop.test(c(resp_treat, resp_control),
          c(n_treat, n_control), correct = FALSE)

# Fisher exact test for a sparse 2 x 2 table
tab <- matrix(c(resp_treat, n_treat - resp_treat,
                 resp_control, n_control - resp_control),
              nrow = 2, byrow = TRUE)
fisher.test(tab)

# Exact normal-theory variance ratio test
var.test(treatment, control)
```

Listing 2: Nonparametric tests and robust spread procedures

```
# One-sample sign test for median = median0
keep <- change != median0
binom.test(sum(change[keep] > median0), sum(keep), p = 0.5)
```

```

# One-sample / paired Wilcoxon signed-rank test
wilcox.test(change, mu = location0, exact = TRUE,
             alternative = "two.sided")
wilcox.test(treatment, control, paired = TRUE, exact = TRUE)

# Monte Carlo one-sample sign-flip test for symmetry about zero
signflip_mean_test <- function(d, B = 9999) {
  obs <- mean(d)
  n <- length(d)
  signs <- matrix(sample(c(-1, 1), B * n, replace = TRUE), nrow = B)
  t_star <- drop(signs %*% abs(d)) / n
  p <- (1 + sum(abs(t_star) >= abs(obs))) / (B + 1)
  list(estimate = obs, p.value = p, null.statistics = t_star)
}

# Independent-sample rank-sum and KS tests
wilcox.test(treatment, control, exact = FALSE)
ks.test(treatment, control)

# Fligner-Killeen rank-based test of scale
fligner.test(list(treatment, control))

# Brown-Forsythe: ANOVA on absolute deviations from group medians
group <- factor(c(rep("treat", length(treatment)),
                  rep("control", length(control))))
value <- c(treatment, control)
center <- ave(value, group, FUN = median)
deviation <- abs(value - center)
summary(aov(deviation ~ group))

```

Listing 3: Null-centered bootstrap for a one-sample mean

```

boot_mean_test <- function(x, mu0, B = 9999) {
  n <- length(x)
  x0 <- x - mean(x) + mu0      # impose H0
  t_obs <- (mean(x) - mu0) / (sd(x) / sqrt(n))
  t_star <- replicate(B, {
    xb <- sample(x0, n, replace = TRUE)
    (mean(xb) - mu0) / (sd(xb) / sqrt(n))
  })
  p <- (1 + sum(abs(t_star) >= abs(t_obs))) / (B + 1)
  list(p.value = p, null.statistics = t_star)
}

# Paired bootstrap: pass treatment - control differences.
# Two-sample bootstrap: center within arms to impose equal means,
# resample independently within each arm, then recompute a studentized
# mean difference. Resample patient pairs/vectors together when paired.

```

Listing 4: Randomization test for a completely randomized trial

```

randomization_test <- function(y, treated, B = 9999) {
  # one outcome per randomized patient; fixed treatment count

```

```

obs <- mean(y[treated]) - mean(y[!treated])
n_treat <- sum(treated)
null_stats <- replicate(B, {
  z <- rep(FALSE, length(y))
  z[sample.int(length(y), n_treat)] <- TRUE
  mean(y[z]) - mean(y[!z])
})
p <- (1 + sum(abs(null_stats) >= abs(obs))) / (B + 1)
list(estimate = obs, p.value = p, null.statistics = null_stats)
}

```

5 Method-selection checklist

1. **Name the estimand.** Mean, proportion, variance/scale, median, location shift, probabilistic superiority, whole-distribution equality, or a multivariate contrast?
2. **Respect the design.** Independent patients, matched pairs, repeated measures, randomized assignment, clustering, and missingness imply different valid resampling/calibration schemes.
3. **State the null.** For example, equal means is weaker than equal distributions; zero average treatment effect is weaker than the sharp no-effect null.
4. **Choose a statistic for that null.** A t statistic targets a mean contrast; a rank-sum statistic targets a rank/probabilistic contrast under general distributions; KS targets the CDF; F targets a variance ratio under normality.
5. **Justify calibration.** Exact model, asymptotic limit, bootstrap null construction, exchangeability, or actual randomization design.
6. **Report the result.** Effect estimate and uncertainty, statistic, reference law or resampling scheme, p-value, sample sizes, and the assumptions that determine interpretation.

Summary

Classical tests are not interchangeable labels. The one-sample t test, signed-rank test, and sign test can all analyze a single sample, but test different hypotheses and rely on different assumptions. Likewise, a pooled two-sample permutation test, a Welch test, and a randomization test can use similar-looking statistics while testing different nulls. A defensible analysis states the estimand, null, sampling design, statistic, calibration, and scope of the conclusion.